

CHAPTER 1

**Tang
&
Yang**

Rong
Tang

Yang
Yun

2019–
2025

An-
nals
of
Sta-
tis-
tics
JRSS
B
AIS-
TATS
ICML

MCMC

1.

1.1.

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D

$$\frac{d}{D} \ll$$

D

GAN
VAE

(1)

(2)

(3)

MCMC

Tang-
Yang

1.1.

Tang
&
Yang
2019–
2025

8

MCMC

1

2
VAE

3

MCMC

2.

2.1.

•

$\mathcal{M}^*$

$\mathbb{R}^D$

$$d$$

$$\beta-$$

$$\beta \geq 1$$

$$P^*$$

$$\mathcal{M}^*$$

$$\mathrm{vol}_{\mathcal{M}^*}$$

$$p^*$$

$$\alpha-$$

$$\alpha \in [0,\beta-1]$$

p^*
 $\mathbb{R}^D$

Lebesgue

Lebesgue

 Par-
ti-
tion

of
Unity
Tang

$$(U_\lambda, \varphi_\lambda)$$

$$\{\gamma_\lambda\}$$

Tang
(2022)

Tang
(2023a)

Reach

reach

$$\text{reach}(\mathcal{M}) = \inf\{r > 0 : \mathcal{M}$$

r -

}

2.2.

IPM.

KL

Hellinger

Tang

IPM

$$d_{\mathcal{F}}(\mu,\nu)=\sup_{f\in\mathcal{F}}\left|\int_{\mathbb{R}^D}f(x)\,\mathrm{d}\mu-\int_{\mathbb{R}^D}f(x)\,\mathrm{d}\nu\right|.$$

$\mathcal{F}$

γ -
Hölder

Sobolev

1-
Lipschitz

Wasserstein-
1

RKHS

MMD
 γ

trade-
off

$\gamma =$
1

d_γ

Wasserstein-
1

$\gamma \rightarrow$
 0^+

d_γ

Tang
(2025a)

2.3.

$\cdot$
Tang

$$= n^{-1/2} \vee n^{-\frac{\alpha+\gamma}{2\alpha+d}} \vee n^{-\frac{\beta\gamma}{d}}.$$

$$n^{-1/2}$$

$$d$$

$$\beta-$$

$$d$$

Haus-
dorff

D

Lep-
ski

chain-
ing
peel-
ing
lo-
cal-
iza-
tion

Gir-
sanov

Poincaré

2.4.

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Gauss-
ian

Langevin

$$p_{\sigma}(x) = \int p_{\text{data}}(y) \phi_{\sigma}(x - y) \, dy$$

$$L_\infty$$

-

$$\nabla \log p_t(x)$$

$$t$$

$$t = 0$$

$$t = T$$

Tang
(2024a)

$\mathcal{S}_{\text{NN}}$
 t

Girsanov

Gir-
sanov

SDE

3.

3.1. Tang
&
Yang
(2022)

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Rong
Tang
and
Yun
Yang.
“Min-
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Rate
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Sub-
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Losses.”
An-
nals
of
Sta-
tis-
tics,
2022.

 D

GAN
VAE

γ

$$n^{-1/2} \vee n^{-\frac{\alpha+\gamma}{2\alpha+d}} \vee n^{-\frac{\beta\gamma}{d}},$$

 α
 β
 d
 D

par-
ti-
tion
of
unity

VAE

Taylor

Lepski

Fano

Varshamov-
Gilbert

Tang
(2022)

Tang
(2023a,
2024a)

3.2. Tang
&
Yang
(2023a)
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Rong
Tang
and
Yun
Yang.

“Es-
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with
Low-
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Struc-
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Gen-
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Mod-
els.”
ICML,
2023.

VAE
GAN

**MLD-
MAE**

K

-

ρ_k

1-
Wasserstein

α -

β -

d

$$O\left(n^{-\frac{\alpha\wedge(\beta-1)+1}{2(\alpha\wedge(\beta-1))+d}} \vee \frac{\log n}{\sqrt{n}}\right),$$

$$\alpha \leq \beta - 1$$

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Tang
(2022)

Tang
(2022)

3.3. Tang
et
al.
(2023b)

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Rong
Tang,

Anir-
ban
Bhat-
tacharya,
Deb-
deep
Pati,
and
Yun
Yang.
“Ro-
bust
Bayesian
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Rie-
mann-
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Sub-
man-
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fold,”
JRSS
B,
2023.

Stiefel

Un-
cer-
tainty
Quan-
tifi-
ca-
tion

**Bayesian
RPE-
TEL**

(1)

(2)

(3)

Bernstein-
von
Mises
(BvM)

set-
ting

Me-
trop-
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lis
(RRWM)

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con-
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pro-
file

3.4. Tang
&
Yang
(2023c)
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Rong
Tang
and
Yun

Yang.
“Min-
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Non-
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Two-
Sample
Test
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Ad-
ver-
sar-
ial
Losses,”
*AIS-
TATS*,
2023.

MMD

$O(n^2)$

Besov

Besov

1

ℓ_1
 ℓ_2

Wasser-
stein

1

2

$$O(n \log n + n^{2d/(4\alpha+d)})$$

MMD

$$O(n^2)$$

**3.5. Tang
&
Yang
(2024a)**

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Rong
Tang
and
Yun
Yang.
“Adap-
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Man-
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fold
Struc-
tures,”
*AIS-
TATS*,
2024.

SOTA

Langevin

—

d

D
Langevin

Gauss-
 ian

$\tilde{O}(\sigma)$

σ

n

d L_∞

—

1-
Wasserstein

$$\frac{1}{\sqrt{n}} \vee n^{-\frac{\alpha+1}{2\alpha+d}}.$$

Tang
(2022)

Tang
(2022)

3.6. Tang (2024b)

Metropolis-
Adjusted
Langevin

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Rong
Tang
and
Yun
Yang.
“Metropolis-
Adjusted
Langevin
Al-
go-
rithm:
Op-
ti-
mal

Di-
men-
sion
De-
pen-
dency
and
Be-
yond.”
ICML,
2024.

MALA

MCMC

$$O(d^{1/3})$$

s-
conductance
pro-
file

MALA

$O(d^{1/3})$

burn-
in

s-
conductance
pro-
file

con-
duc-
tance

Chen
et
al.
(2020)

con-
duc-
tance
pro-
file

warm-
start

$\log M_0$

$\log \log M_0$

**Tang
(2023b)**

Tang
(2023b)

BvM

BvM

MALA

3.7. Tang
&
Yang
(2025a)

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Rong
Tang
and
Yun
Yang.
“Min-
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Rates
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Re-
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on

Man-
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folds
and
Dis-
tri-
bu-
tions,”
arXiv,
2025.

X

Y

$\mu_{Y|X}^*$

X Y

IPM

(1)

Regime

1

Lebesgue

$$n^{-\frac{\alpha_X}{2\alpha_X+d_X}}+n^{-\frac{\alpha_Y+\gamma}{2\alpha_Y+D_Y+\frac{\alpha_Y}{\alpha_X}\cdot d_X}}.$$

(2)

Regime

2

:

$$\mathcal{M}_{Y|X}$$

$$X$$

$$\begin{array}{l} n^{-\gamma\beta_Y/d_Y} \\ (3) \end{array}$$

Regime
3

:

$$\mathcal{M}_{Y|x}$$

$$x$$

$$n^{-\gamma/(\frac{d_Y}{\beta_Y}+\frac{d_X}{\beta_X})}$$

γ -
Hölder
IPM
 d_γ

$$\gamma = 0$$

1-
Wasserstein

$$\gamma = 1$$

3.8. Tang
&
Yang
(2025b)
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Rong
Tang
and
Yun
Yang.
“Con-
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Dif-
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sion
Mod-
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are
Minimax-
Optimal
and
Manifold-
Adaptive,”
arXiv,
2025.

Tang
(2024a)

(1)

$$\alpha_X \in [0, 1]$$

(2)

X

Y

Wasser-
stein

$$(d_X, d_Y)$$

$$(D_X, D_Y)$$

lo-
cal-
iza-
tion
and
peel-
ing

tech-
niques

Gir-
sanov

**Tang
(2024a,
2025a)**

Tang
(2024a)

→
Tang
(2025a)

→

4.

Tang-
Yang

2019–
2025

4.1.

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Tang (2022) $\longrightarrow$ Tang (2023a) MLDMAE $\longrightarrow$ Tang (2025a)

4.2.

VAE

.

VAE (Tang 2021) $\longrightarrow$ Tang (2024a) $\longrightarrow$ Tang (2025b)

Tang
(2024a)

$$t > 0$$

Lebesgue

4.3.

MCMC.

Tang (2023b) BvM + RRWM $\xrightarrow{\text{BvM}}$ Tang (2024b) MALA

BvM

MALA

$$d^{1/3}$$

5.

5.1.

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(1)

2021

Tang
&
Yang,
COLT

VAE

(2)

2022

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Tang
&
Yang,
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Tang
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Yang,
ICML
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MAE

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Tang
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JRSS

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BvM

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2023c

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Tang
&
Yang,
AIS-
TATS

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2024a

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Tang
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AIS-
TATS

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Tang
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Yang,

ICML
MALA

(8) **2025a**

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Tang
&
Yang,
arXiv

(9) **2025b**

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5.2.

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(1)

**Par-
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of
Unity**
Tang
(2022,
2023a)

(2)

γ -
Hölder
IPM

Tang
(2022,
2023c,
2025a)

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Gaussian

+

Tang
(2024a)

Langevin

L_∞

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Tang
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conductance
pro-
file
Tang
(2024b)

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**Bernstein-
von
Mises**

Tang
et
al.
(2023b)

BvM

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ting

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Girsanov

Tang
(2024a,
2025b)

Gir-
sanov

SDE

6.

Tang-
Yang

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Tang
(2022)

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Tang
(2023c)

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1.2.

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